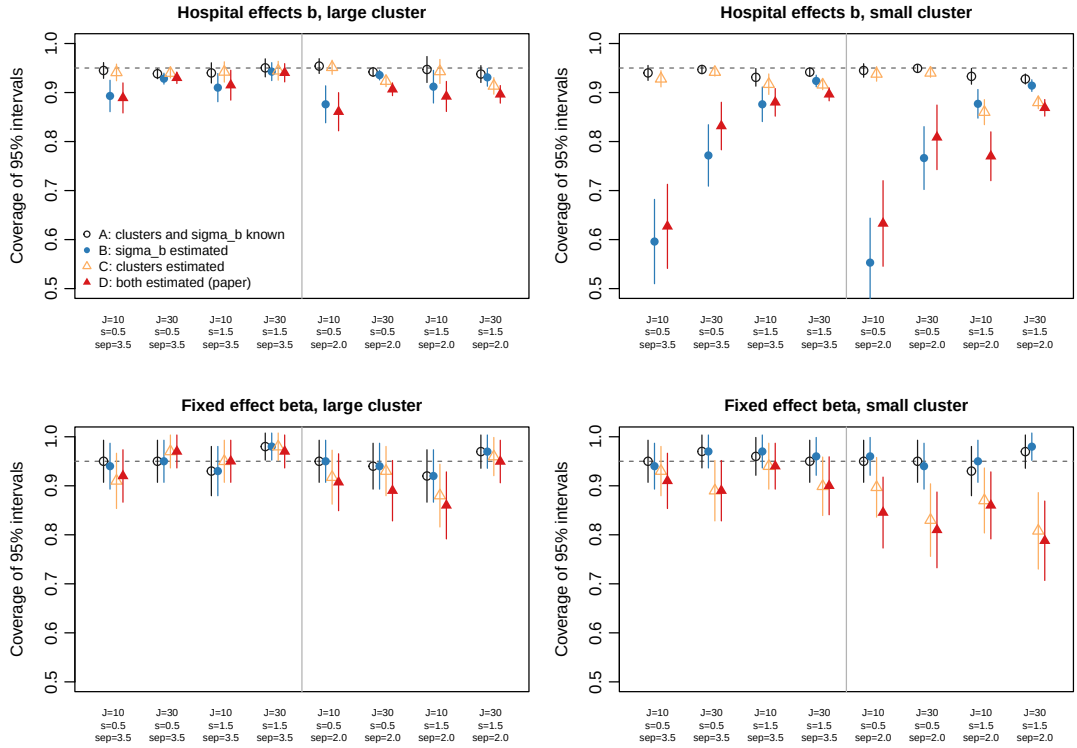
Coverage of Wald and `REsim` intervals in a cluster-weighted mixed-effects logistic model: the cost of estimating σ_b and the cluster assignment

Andrea Gobbo – Summary of a simulation study

Aim: in Caldera et al. (*Biometrics*, 2025) the ML-CWMd is used to predict patient mortality and to estimate the random effect of each hospital on it: patients are grouped into clusters with a classification EM and, within each cluster, mortality follows a logistic regression with a hospital random intercept, $\text{logit}\, P(y_{ij}=1) = \alpha_c + \beta_c x_{ij} + b_{j,c}$, $b_{j,c} \sim N(0, \sigma^2_{b,c})$, fitted with `lme4::glmer`. The uncertainty of the fixed effects β_c is given by Wald intervals and that of the hospital effects $b_{j,c}$ by `merTools::REsim` intervals. Both are computed on the final cluster assignment, treated as known; the `REsim` intervals also fix $\sigma_{b,c}$ at its estimate. This study measures, by simulation, how the coverage of these intervals (share containing the true value, nominally 0.95) changes when the assignment and σ_b are estimated rather than known, separating the two contributions.

Design: on each simulated dataset the regression of each cluster is fitted four times: **A** true clusters, σ_b known; **B** true clusters, σ_b estimated; **C** clusters estimated by CEM, σ_b known; **D** clusters estimated by CEM, σ_b estimated (as in the paper). The datasets comprise J hospitals with 70 patients each; each patient independently belongs to the large cluster with probability fixed at 0.85, so each hospital has about 60 patients in the large cluster and 10 in the small one; a normal clustering covariate with mean distance = sep, one binary regressor with a different β in each cluster. Scenarios: $J \in \{10, 30\}$ hospitals, $\sigma_b \in \{0.5, 1.5\}$, sep $\in \{3.5, 2\}$; 100 replicates each. Hospital intervals are computed with `REsim` as in the original code. With σ_b known, the `lme4` Laplace likelihood is minimised over the fixed effects only. Monte Carlo standard errors of the coverage, across the 100 replicates: 0.004–0.046.



Coverage by scenario, cluster and version; bars: ±1.96 Monte Carlo standard errors; left half of each panel sep = 3.5, right half sep = 2.

Results.

- *Check.* With assignment and σ_b known (A), coverage is 0.92–0.98 for β and 0.93–0.95 for the hospital effects.
- *Hospital effects: the main factor is $\hat{\sigma}_b$.* In the small cluster with $\sigma_b = 0.5$, estimating σ_b gives coverage 0.55–0.60 with $J = 10$ and 0.77 with $J = 30$ (B), since $\hat{\sigma}_b = 0$ in 30–39% and 9–11% of the replicates, in which case all intervals have zero width. Estimating the assignment alone has little effect (C, 0.93–0.94). With $\sigma_b = 1.5$, B covers 0.88–0.92; in the large cluster all versions cover 0.86–0.95.
- *Fixed effects: the main factor is the assignment.* A and B differ by at most 0.02: the Wald SEs of `glmer` already include the uncertainty of $\hat{\sigma}_b$. In the small cluster with $J = 30$, C and D cover 0.89–0.90 with sep = 3.5 and 0.79–0.83 with sep = 2; the estimated small cluster contains 8–9% and 14–26% of patients from the other cluster.

Limitations. Simplified model (two clusters, one clustering covariate, one regressor) and CEM (one k-means initialisation); results depend on how much β differs between clusters.